

GeCCo: AI as the New Scientist

How Large Language Models are Decoding the Human Mind

 AI brain network

The Bottleneck in Science

Traditionally, scientists build "Models of the Mind" by hand. This process is:

- **Slow:** Taking months of manual coding and testing.
- **Complex:** Requiring years of expert training.
- **Biased:** Humans often stick to theories they already like.



Jargon Buster: What is a Model?

The "Brain Recipe"

A cognitive model is like a computer program that acts as a **blueprint** for how people think. It's a mathematical recipe for behavior.

Predicting the Future

By fitting these recipes to data, we can predict what a person will do next when faced with a choice or a memory task.

Theory Testing

We use models to see if our theories (like "people have bad memory under stress") actually match the real numbers.

Computational Science

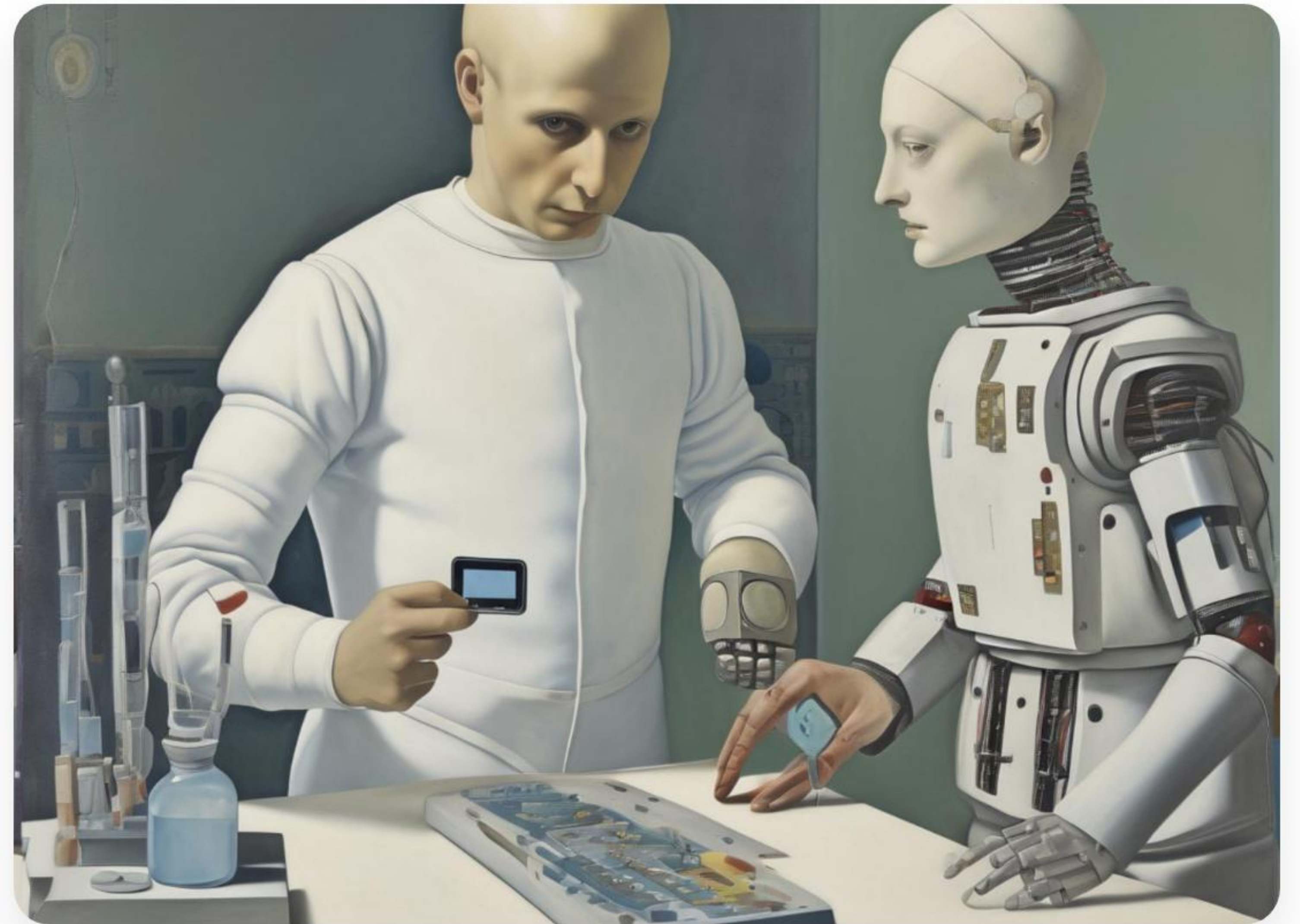
This allows us to turn vague psychological ideas into **hard science** that computers can understand.

Introducing GeCCo

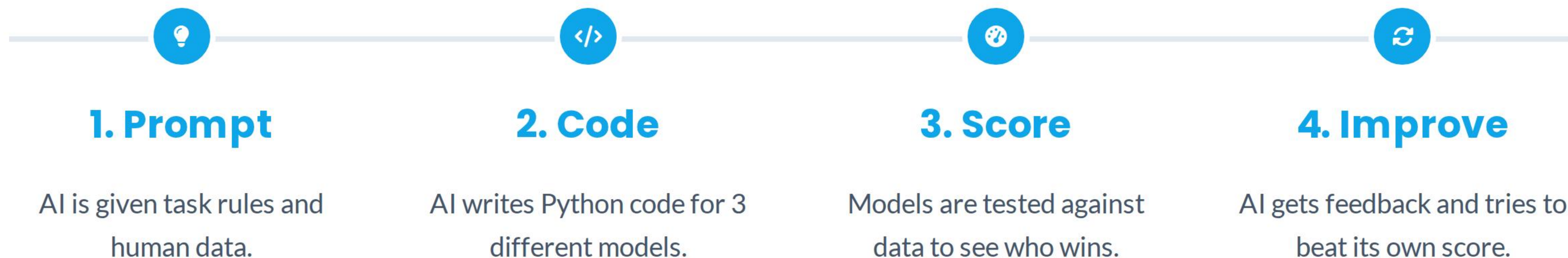
The AI Researcher

The paper introduces **GeCCo** (Guided generation of Computational Cognitive Models).

Instead of a human writing the math, an AI acts as the creative scientist—brainstorming, coding, and testing its own theories about the mind.



The Loop of Discovery



The AI's Toolkit



LLMs

Large Language Models (like Llama 3 or ChatGPT). They have read millions of books and know how humans think and code.



BIC

Bayesian Information Criterion. Think of this as an "Efficiency Score." It rewards accuracy but punishes complexity.

$$\text{BIC} = k \cdot \ln(n) - 2 \cdot \ln(\hat{L})$$

Simplified: Score = (Complexity Penalty) - (Accuracy Reward). **Lower is better!**

Result: AI vs. Human Experts

In a standard Decision-Making task, the AI models achieved significantly better scores than the best human-made theories.

Human Model (pWADD)

52.0

AI Model (GeCCo)

39.4

Lower BIC means the AI found a much more "elegant" explanation for the same data.

Winning Across the Board

99%

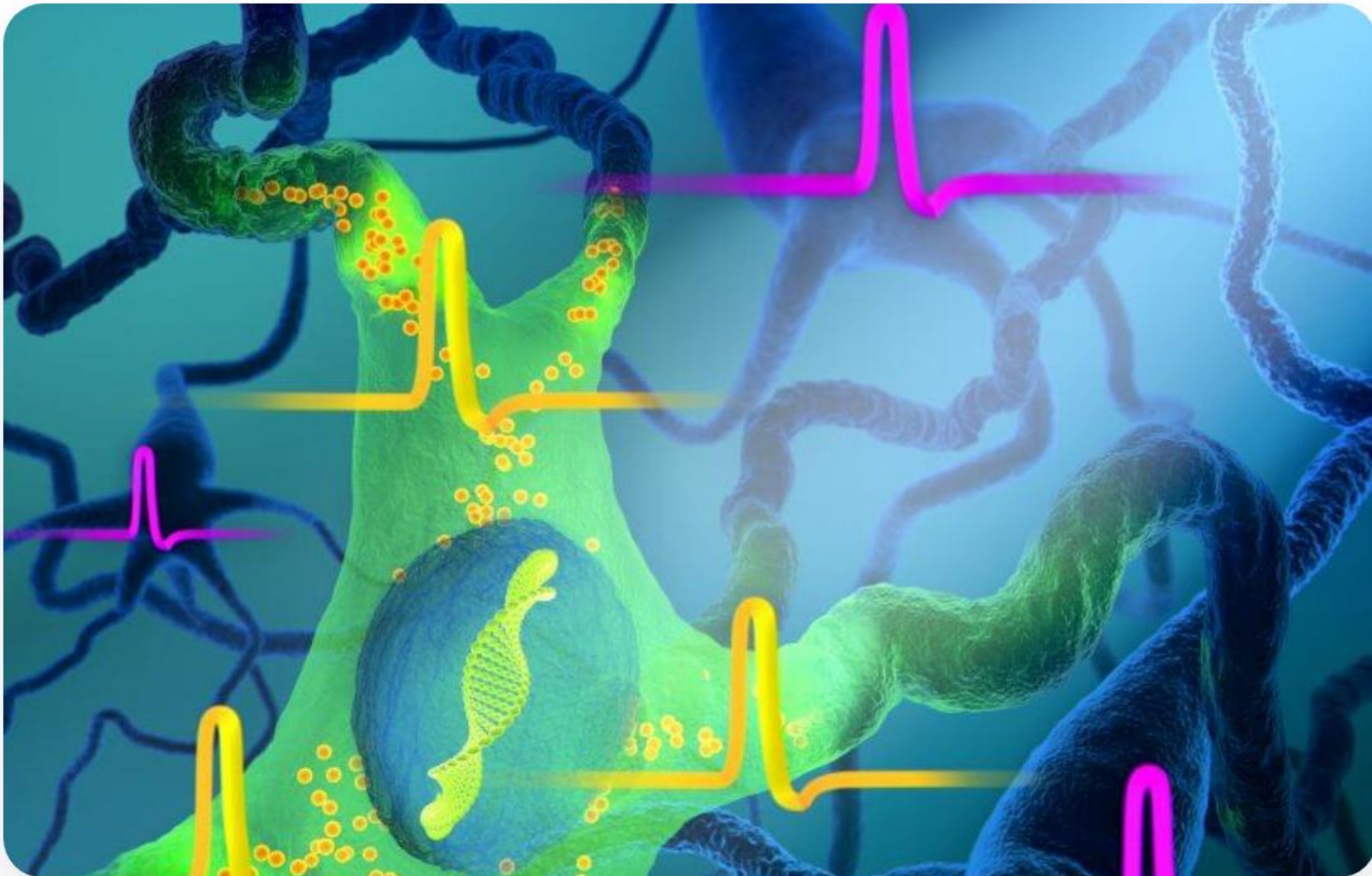
Confidence Score

Beyond Decision Making

The AI wasn't just lucky once. It outperformed humans in:

- ✓ **Learning:** How we learn from rewards.
- ✓ **Planning:** How we think steps ahead.
- ✓ **Memory:** How our "scratchpad" brain works.

Is the AI Cheating?



Scientists ran "Control Experiments" to be sure:

🛡️ **No Memorization:** Tests showed the AI hadn't seen the data before. It was genuinely reasoning.

📊 **Ground Truth:** When given fake data with a "secret rule," the AI successfully worked backward to find it.

🔗 **Interpretability:** Unlike most AI, GeCCo outputs clean code that humans can read and learn from.

The Critic's Corner

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"While GeCCo is faster, could it lead scientists to rely too much on AI and lose their creative intuition? There is also a risk that the AI might inherit biases from its training data."

Challenge 1

Dependence on starting templates.

Challenge 2

Complexity of real-world "messy" data.

A New Era for Research

GeCCo proves that AI isn't just for chatting—it's a tool that can democratize science, making complex discovery available to everyone, and accelerating our understanding of the human mind.

Questions?

Thank you for your attention!

Presentation on "Guided generation of Computational Cognitive Models"

Based on the research of MilenaCCNlab

Image Sources



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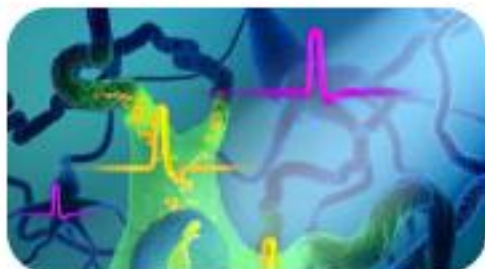
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